

ASSIGNMENT REPORT

“Recognizing the Ethical and Societal Implications of Engineering Solutions Created Using Artificial Intelligence and Machine Learning”

**Applications, Intelligent Agents, PEAS,
Societal Impact and Ethical Concerns**

Selected Applications:

- i. Facial Recognition & AI Surveillance
- ii. AI-Based Recruitment / Hiring

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Abstract

This report examines the ethical and societal implications of Artificial Intelligence (AI) engineering solutions, with focused analysis on two high-impact applications: Facial Recognition and AI Surveillance systems, and AI-Based Recruitment and Hiring platforms. Employing the PEAS (Performance Measure, Environment, Actuators, Sensors) framework as its analytical foundation, the report systematically characterises each application across five types of intelligent agents—Simple Reflex, Model-Based Reflex, Goal-Based, Utility-Based, and Learning agents—and five classes of operational environments, generating twenty-five possible agent-environment combinations, of which ten are examined in depth for each application.

The analysis reveals that the ethical implications of AI deployment are not merely incidental to technical design choices—they are constitutive of them. The choice of agent architecture, utility function, observability assumptions, and learning mechanism each carries profound consequences for fairness, accountability, privacy, and human dignity. The report further examines the legal and ethical challenges AI poses to contemporary society, with particular attention to privacy and surveillance, algorithmic bias and discrimination, and the appropriate role of human judgment in automated decision-making. The report concludes that responsible AI engineering requires the integration of ethical reasoning throughout the design and deployment lifecycle, supported by robust regulatory frameworks and genuine democratic accountability.

1. Introduction to Artificial Intelligence

Artificial Intelligence (AI) is broadly defined as the simulation of human intelligence processes by machine systems, encompassing capabilities such as learning from experience, reasoning under uncertainty, perception of the environment, natural language understanding, and goal-directed problem-solving. From its theoretical foundations in the mid-twentieth century to its contemporary deployment in billions of daily interactions, AI has undergone a transformation from a research curiosity to a pervasive societal infrastructure.

As AI systems are increasingly embedded in critical domains—from healthcare diagnostics and criminal justice risk assessment to financial credit scoring, educational evaluation, and national security—the engineering community bears a heightened responsibility. The design of an AI system is not a value-neutral technical act; it is a moral and political intervention that shapes the distribution of opportunities, risks, and powers within society. Understanding this is the foundational premise of this report.

The two applications analysed in this report—Facial Recognition and AI Surveillance, and AI-Based Recruitment and Hiring—represent cases in which the stakes of algorithmic decision-making are extraordinarily high. In both cases, the systems exercise power over fundamental human interests: the freedom to move through public space without being tracked, and the opportunity to participate in economic life through fair access to employment. In both cases, the technical sophistication of the systems creates a false appearance of objectivity that can obscure deeply embedded biases and structural inequities.

This report proceeds through a structured examination of each application's intelligent agent architecture using the PEAS framework, followed by detailed assessment of ten agent-environment combinations per application, and a comprehensive evaluation of the legal, ethical, and societal dimensions. The analysis is situated within the broader context of AI ethics scholarship and regulatory developments.

2. Overview of AI Applications and Their Categories

2.1 Categories of Artificial Intelligence

AI systems are conventionally classified along a capability spectrum with three tiers:

- **Narrow AI (Weak AI):** Systems designed and trained to perform specific, well-defined tasks. All commercially deployed AI systems currently fall within this category—facial recognition systems, language models, recommendation engines, and hiring tools are all examples of narrow AI operating within strictly bounded domains.
- **General AI (Strong AI):** Hypothetical systems capable of performing any intellectual task that a human can perform. No such system exists; this remains a research aspiration.
- **Artificial Superintelligence (ASI):** A theoretical system capable of surpassing human cognitive performance across all domains. Purely speculative at present.

Within the narrow AI category, applications are further classified by their primary cognitive modality:

- **Machine Learning (ML):** Systems that improve performance on tasks through exposure to data, without being explicitly programmed for each scenario. ML encompasses supervised, unsupervised, and reinforcement learning paradigms.
- **Deep Learning:** A subset of ML employing multi-layered artificial neural networks, particularly effective for perceptual tasks such as image recognition, speech processing, and natural language understanding.
- **Natural Language Processing (NLP):** AI systems designed to understand, generate, and reason about human language—powering resume parsing, chatbots, sentiment analysis, and machine translation.
- **Computer Vision:** AI systems that interpret and derive meaning from visual information—the foundational technology underlying facial recognition and surveillance applications.
- **Expert Systems:** Rule-based systems encoding domain expertise in structured condition-action logic, representing the earliest deployed AI paradigm.

2.2 The Selected Applications in Context

Facial Recognition and AI Surveillance systems represent a convergence of Computer Vision, Deep Learning, and real-time data processing. Their deployment in public spaces has expanded dramatically over the past decade, driven by reductions in computational cost and improvements in recognition accuracy. They exemplify AI applications where the potential for social benefit (public safety, missing person identification) coexists with profound potential for social harm (mass surveillance, discriminatory targeting, political suppression).

AI-Based Recruitment and Hiring systems represent a convergence of NLP, ML, and psychometric evaluation. They are deployed by organisations across all sectors to manage the growing volume of job applications, reduce time-to-hire, and—ostensibly—improve objectivity in candidate evaluation. They exemplify AI applications where the promise of efficiency and standardisation obscures the reproduction and amplification of historical inequities.

3. Types of Intelligent Agents

An intelligent agent is any entity that perceives its environment through sensors and acts upon that environment through actuators. Agents are characterised by their architecture: the internal mechanisms that determine how they translate percepts into actions. The AIMA (Russell & Norvig, 2021) framework identifies five progressively sophisticated agent types:

Agent Type	Core Mechanism	Suitable For	Key Limitation
Simple Reflex Agent (SRA)	Condition-action rules on current percept only	Fully observable, deterministic, simple environments	Cannot handle partial observability or uncertainty
Model-Based Reflex Agent (MBRA)	Maintains internal state/model of world; uses percept history	Partially observable environments requiring state tracking	Model may be inaccurate; computationally heavier
Goal-Based Agent (GBA)	Selects actions that achieve a specified goal through planning	Complex environments requiring multi-step reasoning	Goal specification is difficult; goals may conflict with ethics
Utility-Based Agent (UBA)	Selects actions maximising expected utility across outcomes	Environments with competing objectives and uncertainty	Utility functions embed values; optimisation can have unintended consequences
Learning Agent (LA)	Learns from experience; improves performance via feedback	Complex, dynamic, non-stationary environments	Can learn and amplify biases; requires careful oversight

The choice of agent architecture is not merely a technical decision—it reflects assumptions about the nature of the environment, the stability of objectives, and the acceptable degree of autonomous decision-making. As agent sophistication increases, so too does both the capability and the potential for consequential harm.

4. Classification of AI Environments

The operational environment of an AI agent fundamentally shapes both the appropriate choice of agent architecture and the ethical implications of its deployment. Environments are characterised along five independent dimensions (Russell & Norvig, 2021):

Dimension	Classification	Description
1. Observability	Fully Observable	All information relevant to the agent's decision is accessible through its sensors at any point in time. No hidden state exists.
	Partially Observable	Only a subset of the relevant information is accessible. The agent must reason about hidden or inferred states.
2. Determinism	Deterministic	The next state of the environment is completely determined by the current state and the agent's action. No randomness.
	Stochastic	Uncertainty and randomness affect outcomes. The same action in the same state may produce different results.
	Strategic	The environment is affected by the actions of multiple agents, each pursuing their own objectives.
3. Episodicity	Episodic	Each decision episode is independent of prior episodes. The agent's choice in one episode does not affect future episodes.
	Sequential	Current actions affect future states and available actions. The agent must plan across time.
4. Dynamism	Static	The environment does not change while the agent is deliberating. Time pressure is absent.
	Dynamic	The environment can change independently of the agent's actions while the agent is deciding, creating time pressure.
5. Continuity	Discrete	Percepts and actions take values from a finite, countable set. The state space is manageable.

Dimension	Classification	Description
	Continuous	Percepts and/or actions can take any value within a continuous range. The state-action space is infinite-dimensional.

Real-world AI deployments almost always occupy the more challenging end of each dimension: partially observable, stochastic, sequential, dynamic, and continuous. This environmental complexity is a primary driver of both the difficulty of building effective AI systems and the ethical risks they pose—since the more complex the environment, the harder it is to predict, explain, or audit system behaviour.

5. Application 1: Facial Recognition and AI Surveillance

5.1 System Overview

Facial recognition technology (FRT) is a biometric AI application that identifies or verifies individuals by analysing distinctive geometrical and textural features of the human face. Modern FRT systems employ deep convolutional neural networks (CNNs), typically trained on datasets of millions of facial images, to extract high-dimensional feature embeddings that can be compared across individuals with high accuracy under controlled conditions.

AI surveillance systems integrate FRT with real-time video processing infrastructure, automated alert mechanisms, and—increasingly—behavioural analytics capabilities such as gait recognition, emotional inference, and crowd dynamics analysis. These systems are deployed in contexts ranging from access control and airport security to city-wide public surveillance, retail loss prevention, and law enforcement operations.

The global facial recognition market was valued at approximately USD 5.7 billion in 2024 and is projected to exceed USD 19.3 billion by 2030. Major deployments include the UK National Police Chiefs' Council's automated facial recognition trials, China's city-wide surveillance networks, and commercial deployments by Amazon Rekognition, Microsoft Azure Face API, and Clearview AI.

5.2 PEAS Representation

The PEAS framework provides a structured specification of the operational context for an AI agent. For a facial recognition and surveillance agent:

PEAS Component	Description
Performance Measure	Recognition accuracy (true positive rate, false positive rate, and false negative rate across demographic groups); real-time processing throughput (frames per second); system availability and uptime; alert precision and recall; speed of confirmed identification; reduction in security incidents; coverage area maintained; demographic parity across racial, gender, and age groups.
Environment	Physical environments: public streets, transport hubs, airports, stadiums, shopping centres, government buildings, corporate campuses; environmental conditions: variable lighting (day/night, indoor/outdoor, artificial/natural), weather (rain, fog, glare), crowd density, camera angle and resolution, distance from subject; digital environment: real-time video streams, biometric reference databases, alert management systems, law enforcement record systems.
Actuators	Visual alert systems (on-screen highlights, bounding boxes); automated alarm triggers (audio, visual, network signals); access control actuators (electronic locks, barrier gates, door release systems); law enforcement notification interfaces; database record creation and update systems; dashboard displays for human operators; blacklist/watchlist management interfaces.
Sensors	High-resolution CCTV cameras (2–8 megapixel, 15–60 fps); infrared and thermal imaging cameras for low-light and night operation; pan-tilt-zoom (PTZ) camera control systems; network interfaces to biometric reference databases; frame capture and pre-processing pipelines; GPU-accelerated inference hardware; GPS/positioning systems for multi-camera coordination; audio sensors for supplementary behavioural cues.

5.3 Agent-Environment Combinations: Analysis of Selected 10 of 25

The following table presents a structured analysis of ten significant agent-environment combinations, drawn from the full matrix of 25 possible pairings (5 agent types $\times$ 5 environment dimensions). Each combination is evaluated for operational characteristics, limitations, and ethical implications specific to facial recognition and surveillance deployment.

#	Agent Type	Environment Characteristic	Analysis & Ethical Implications
C1	Simple Reflex Agent	Observability: Fully Observable	In a controlled surveillance corridor with ideal lighting, the SRA applies deterministic condition-action rules: "IF face_match = TRUE AND confidence $\geq$ 0.85 THEN trigger_alert." All frames are fully captured and processed. The agent reacts instantaneously to each percept without retaining any prior context. Limitation: The SRA has no contextual reasoning, temporal awareness, or intent analysis. It fires the same rule for an authorised employee, a visitor, or a look-alike. Context-free matching in fully observable environments is technically straightforward but operationally shallow and dangerously reductive. Ethical Concern: Treating every match as an actionable event in a fully observed public space places millions of citizens under automated surveillance without consent, proportionality, or meaningful right of appeal. This violates the presumption of innocence and the right to anonymous movement in public space—a cornerstone of liberal democratic societies.
C2	Simple Reflex Agent	Determinism: Stochastic	Real-world surveillance is inherently stochastic: face angles, partial occlusions, variable lighting, rain, and camera quality introduce irreducible randomness. The SRA ignores this uncertainty entirely and applies identical condition-action rules regardless of confidence level or environmental variability. Limitation: The SRA cannot reason under uncertainty. Its binary rules fire or do not, regardless of whether confidence is 90% or 52%. This mismatch between a probabilistic environment and a

#	Agent Type	Environment Characteristic	Analysis & Ethical Implications
			deterministic rule-set produces high false-positive rates—particularly for demographic groups underrepresented in training data. Ethical Concern: Research by Buolamwini & Gebru (2018) demonstrated error rates up to 34.7% higher for darker-skinned females than lighter-skinned males in commercial facial recognition systems. An SRA in a stochastic environment will disproportionately flag minority groups, institutionalising discrimination in public safety infrastructure.
C3	Model-Based Reflex Agent	Observability: Partially Observable	Urban surveillance is never fully observable—cameras have blind spots, crowds create occlusions, and angles vary. The MBRA maintains an internal model of tracked individuals: position estimate, identity confidence score, last-seen timestamp, and inferred trajectory. This model enables reasoning about individuals even when direct percepts are absent. Limitation: The internal model is only as valid as its embedded assumptions. If the model assumes behavioural consistency, it may incorrectly continue tracking the wrong person after a crowd crossing—a "track swap" error that results in wrongful flagging of an innocent person. Ethical Concern: Persistent state maintenance on individuals in partially observable settings enables the construction of detailed long-term profiles without the subject's knowledge or consent. This capability fundamentally undermines informational privacy and enables a form of state power disproportionate to any legitimate security objective.
C4	Model-Based Reflex Agent	Dynamism: Dynamic	Urban environments change constantly: crowds shift, lighting varies across the day, subjects change appearance, and new individuals continuously enter the frame. The MBRA updates its internal state incrementally to reflect these changes, re-estimating identity, position, and threat level.

#	Agent Type	Environment Characteristic	Analysis & Ethical Implications
			Limitation: Rapid dynamic change can corrupt the internal model faster than it is updated. A crowd crossing can cause the system to lose track of one subject and transfer their tracking token to a different person, producing wrongful alerts and consequential police contact for innocent individuals. Ethical Concern: Dynamic tracking across time and space enables construction of behavioural profiles: shopping patterns, social associations, attendance at political meetings or houses of worship. This transforms surveillance from a security instrument into a mechanism of social control, producing chilling effects on constitutionally protected freedoms of assembly and religion.
C5	Goal-Based Agent	Episodicity: Sequential	Surveillance is inherently sequential: face detection triggers an alert; the alert dispatches security; contact creates a record; the record influences all future encounters. The GBA reasons from a defined goal (e.g., "prevent unauthorised access") backward through action sequences to the current decision, selecting the action best positioned to achieve the goal over time. Limitation: Sequential reasoning creates compounding lock-in effects. An incorrect identification at step 1 initiates a cascade of escalating interventions—flagging, approach, questioning, recording—even if the initial match was erroneous. The sequential structure amplifies single errors into systemic harms. Ethical Concern: Errors in sequential surveillance chains have permanent consequences. An incorrect algorithmic match can result in a "person of interest" designation that persists in law enforcement databases across thousands of future interactions. There is no effective mechanism for algorithmic accountability in sequential chains spanning multiple agencies and jurisdictions.

#	Agent Type	Environment Characteristic	Analysis & Ethical Implications
C6	Goal-Based Agent	Determinism: Strategic	Modern surveillance ecosystems involve multiple interacting goal-seeking agents: city CCTV networks, private venue cameras, law enforcement biometric databases, immigration systems, and commercial data brokers. Each agent pursues its own objective; the actions of one constrain and enable the options of others, creating a multi-agent strategic environment. Limitation: Strategic multi-agent goal pursuit creates surveillance creep: each agent legitimately pursues its assigned goal, but their interactions enable aggregate capabilities not sanctioned by any single authority. A retail theft-prevention camera network becomes an input to a law enforcement facial recognition system without any explicit authorisation for this expanded scope. Ethical Concern: The absence of a unified ethical governance framework across competing strategic actors—commercial, governmental, and foreign—creates conditions for arbitrary, unaccountable surveillance. No single actor bears responsibility for aggregate system behaviour, rendering democratic oversight practically impossible across the system as a whole.
C7	Utility-Based Agent	Continuity: Continuous	Raw video streams provide continuously valued percepts: pixel intensities, facial landmark coordinate vectors, Euclidean distances between feature points, and match probability distributions. The UBA evaluates all feasible responses against a utility function (e.g., maximising threat detection rate while minimising false alarm rate) and selects the highest-utility action from a continuous action space, typically involving variable confidence thresholds and alert escalation levels. Limitation: The utility function is a value judgment encoded as mathematics. A function that weights detection accuracy heavily at the expense of false-alarm rate will systematically prefer over-flagging. In a continuous action space, small changes to utility

#	Agent Type	Environment Characteristic	Analysis & Ethical Implications
			weights can produce large, non-intuitive shifts in system behaviour—and these weight choices are invisible to the public. Ethical Concern: In continuous environments, utility maximisation can justify mass population surveillance. As the marginal computational cost of scanning one additional face approaches zero and the marginal expected utility of detecting one additional threat appears constant, the utility-maximising action is always to expand surveillance—an outcome structurally incompatible with a free society.
C8	Utility-Based Agent	Observability: Partially Observable	Under partial observability, the UBA computes expected utility across all plausible environmental states, weighting each by its estimated probability and selecting the action maximising total expected utility. When scene information is incomplete, this typically means choosing actions that reduce uncertainty—deploying additional cameras, lowering confidence thresholds, or broadening monitoring scope. Limitation: Utility optimisation under uncertainty creates a structural ratchet effect: when utility is defined primarily in security terms and civil liberties considerations are underweighted or absent from the utility function, the rational response to any uncertainty is always to increase surveillance intensity—never to reduce it. Ethical Concern: This ratchet effect embeds a structural bias toward totalitarian surveillance architectures. The precautionary principle—"we cannot determine who is dangerous, so we must monitor everyone"—becomes a mathematically rational utility-maximising strategy, providing algorithmic justification for the elimination of genuinely private space in public life.
C9	Learning Agent	Dynamism: Dynamic	Learning agents are uniquely suited to dynamic surveillance environments, continuously adapting recognition models to new face presentations, disguise attempts, ageing, environmental variations,

#	Agent Type	Environment Characteristic	Analysis & Ethical Implications
			and emerging threat profiles. The learning component receives feedback from confirmed identifications and refines its model parameters accordingly, improving accuracy over time. Limitation: Unconstrained learning in dynamic environments allows the system to evolve in unintended and fundamentally opaque directions. The agent may learn to associate innocuous behavioural signals—wearing a hood, looking downward, moving with urgency—with threat indicators based on spurious correlations in historical training data, without any human understanding of the causal logic underlying these associations. Ethical Concern: A learning agent without ethical constraints in a dynamic surveillance environment constitutes a direct threat to freedom of movement and association. If the agent learns that attendance at protests or religious gatherings correlates with prior police contact—tautologically, because surveillance is concentrated at these sites—it will flag future attendees, creating a self-fulfilling prophecy of criminalisation.
C10	Learning Agent	Continuity: Continuous	The learning agent processes continuous video streams, refining facial recognition and behavioural analysis models in near-real-time based on a continuous feedback loop between recognition attempts and confirmed identifications. This architecture enables the highest achievable recognition accuracy across diverse and challenging conditions. Limitation: Without explicit debiasing mechanisms, the agent learns and encodes the demographic patterns embedded in its feedback data. As confirmed identifications disproportionately involve certain demographic groups—due to pre-existing patterns in policing—the learning model amplifies these biases with each iteration, while its outputs appear increasingly authoritative.

#	Agent Type	Environment Characteristic	Analysis & Ethical Implications
			Ethical Concern: The combination of continuous learning and continuous data streams creates conditions for an autonomous surveillance apparatus that evolves beyond human comprehension and oversight. The system becomes an epistemic black box whose decision logic is interpretable by no human—a fundamental challenge to democratic accountability, the rule of law, and individual rights that no post-hoc audit can adequately address.

5.4 Societal Impact

The deployment of AI-powered facial recognition and surveillance systems generates profound and multidimensional societal impacts across a range of domains:

Public Safety and Security: Proponents of FRT argue that it offers legitimate public safety benefits, including the ability to identify suspects in real time, locate missing persons, and deter criminal activity through its deterrent effect. London's Metropolitan Police reported a number of successful identifications of wanted individuals through FRT trials. However, the security benefits must be weighed against the systemic costs borne by those subject to wrongful identification.

Chilling Effects on Civil Society: The knowledge—or even the perception—of ubiquitous facial recognition surveillance demonstrably affects individual behaviour. Studies in jurisdictions with pervasive surveillance report reduced participation in public protests, religious gatherings, and community organising. This chilling effect represents a measurable harm to democratic participation, even among individuals who are never directly subject to wrongful identification.

Differential Harm Across Social Groups: The harms of facial recognition systems are not distributed equally. Buolamwini and Gebru's (2018) landmark Gender Shades study demonstrated accuracy disparities of up to 34.7 percentage points between demographic subgroups in commercial FRT systems. This means that the same system that is highly accurate

for lighter-skinned males is dramatically less accurate for darker-skinned females—generating false-positive alerts at dramatically higher rates for already-marginalised communities.

Power Asymmetries: Facial recognition concentrates unprecedented power in the hands of whoever operates the surveillance infrastructure. States can monitor political opposition; corporations can track consumer behaviour; landlords can monitor tenants; intimate partners can surveil one another. The technology transfers power from the observed to the observer in ways that fundamentally alter the dynamics of social control.

International and Geopolitical Dimensions: Authoritarian governments have deployed facial recognition as a tool of political repression against ethnic minorities—most notably China's surveillance of Uyghur populations in Xinjiang. The global diffusion of surveillance technology developed under authoritarian contexts raises fundamental questions about export controls, corporate responsibility, and the use of AI as a vector for human rights violations.

5.5 Ethical Concerns

- **Consent and Informational Privacy:** FRT systems operating in public spaces capture biometric data from individuals who have neither consented to data collection nor been informed that it is occurring. Biometric data is inherently personal and irreplaceable—unlike passwords, faces cannot be changed following a data breach.
- **Algorithmic Bias and Discriminatory Accuracy:** The demonstrated accuracy disparities across demographic groups mean that FRT systems impose systematically higher burdens of wrongful identification on minority communities, amplifying existing patterns of over-policing and racial discrimination.
- **Lack of Transparency and Explainability:** Deep learning FRT systems are fundamentally opaque—they cannot explain the basis of their identification decisions in human-comprehensible terms. This opacity makes legal challenge, democratic accountability, and meaningful oversight practically impossible.
- **Function Creep and Mission Expansion:** Systems deployed for one purpose (airport security) are routinely repurposed for others (tracking political activists, immigration enforcement) without public consultation or legal authorisation. The architecture of surveillance expands to fill the available space.
- **Due Process and the Right to Challenge:** Individuals wrongfully identified by FRT systems have limited practical capacity to discover, challenge, or seek redress for

erroneous identifications, particularly when those identifications are embedded in law enforcement decision chains.

6. Application 2: AI-Based Recruitment and Hiring

6.1 System Overview

AI-based recruitment and hiring (AIHR) systems represent the application of machine learning, natural language processing, and predictive analytics to the candidate screening, evaluation, and selection processes. These systems range in sophistication from simple keyword-matching Applicant Tracking Systems (ATS) to complex multi-modal evaluation platforms that analyse written applications, video interview performance, psychometric assessments, and social media profiles to generate predictive candidate-success scores.

AIHR systems are motivated by the practical challenge of managing high-volume recruitment: large organisations routinely receive thousands of applications for a single position. The promise of AI in this context is threefold: efficiency (reducing human screening time), consistency (applying the same criteria to every candidate), and—most contentiously—objectivity (removing human bias from hiring decisions). It is precisely this last claim that requires the most critical scrutiny.

Prominent examples include HireVue's video interview analysis platform (which was used by major corporations including Unilever and Goldman Sachs before partially discontinuing face analysis following criticism), Pymetrics' neuroscience-based game assessments, LinkedIn Talent Insights, Workday's predictive hiring tools, and the now-infamous Amazon internal AI recruiting tool that was abandoned in 2018 after it was discovered to systematically penalise applications from women.

6.2 PEAS Representation

PEAS Component	Description
Performance Measure	Quality of shortlisted candidates (predictive validity of success metrics); time-to-hire and cost-per-hire reduction; shortlist diversity metrics (demographic representation relative to candidate pool); false positive rate (unsuitable candidates advanced) and false negative rate (qualified candidates rejected); candidate experience ratings; compliance with equal employment opportunity regulations; long-term retention and performance of hired candidates.
Environment	Digital environments: Applicant Tracking Systems (ATS), resume/CV databases, job portals (LinkedIn, Indeed, Glassdoor), social media profiles (LinkedIn, GitHub), video interview platforms, psychometric assessment portals, background check APIs; data environments: resume text, cover letters, structured questionnaire responses, video interview recordings, assessment game outcomes, reference reports; organisational contexts: company culture data, historical performance of prior hires, existing workforce demographic profiles.
Actuators	Automated candidate ranking and shortlisting outputs; rejection and advancement notification emails; interview scheduling interfaces; structured feedback reports for human reviewers; diversity dashboard displays; bias audit reports; recruitment pipeline analytics; API outputs to downstream HR systems; offer letter generation systems.
Sensors	Text parsing and NLP engines for resume and cover letter analysis; optical character recognition (OCR) for document processing; video capture and analysis systems for interview evaluation (facial expression, vocal tone, speech rate, eye contact); psychometric game interfaces and response capture systems; background check API integrations; LinkedIn/job portal data ingestion interfaces; performance database connections for training feedback.

6.3 Agent-Environment Combinations: Analysis of Selected 10 of 25

The following table presents a structured analysis of ten significant agent-environment combinations from the 25-combination matrix for AI-based recruitment and hiring applications. Each combination is evaluated for operational characteristics, limitations, and ethical implications.

#	Agent Type	Environment Characteristic	Analysis & Ethical Implications
C1	Simple Reflex Agent	Episodicity: Episodic	Each job application is treated as a fully independent episode. The SRA applies fixed rule-sets: "IF degree $\in$ {required_list} AND keyword_count $\geq$ threshold AND GPA $\geq$ 3.5 THEN shortlist." Each application is evaluated in isolation against these criteria without reference to the broader candidate pool, prior rounds, or accumulated learning. Limitation: The SRA cannot learn from accumulated screening patterns, adapt to evolving job market realities, or detect that its rules systematically exclude qualified candidates. Once established, the rule-set operates indefinitely without self-correction or human review unless explicitly modified. Ethical Concern: Fixed-rule episodic screening systematically excludes candidates who are equally or more qualified but express their qualifications in non-standard forms—those from non-elite institutions, career changers, and individuals with non-linear career trajectories. The rules encode the assumptions of their authors, which are rarely demographically neutral.
C2	Simple Reflex Agent	Observability: Fully Observable	The SRA has access to all submitted application components: resume, cover letter, portfolio links, assessment results, and references. It applies its fixed condition-action rules to this complete information set, processing each data element independently according to predefined criteria. Limitation: Full observability provides rich data that the SRA cannot meaningfully exploit. The additional information—cover letter tone, portfolio aesthetic,

#	Agent Type	Environment Characteristic	Analysis & Ethical Implications
			email domain, university name—may inadvertently serve as proxies for demographic characteristics that the system was never designed to evaluate, embedding unintended discrimination into a nominally comprehensive process. Ethical Concern: Full data access in an SRA hiring context is counterproductive from an equity perspective. The more data available to a rule-based system, the more opportunities exist for sensitive demographic proxies to contaminate decisions. An SRA with fully observable data may achieve disparate impact discrimination against protected groups while technically applying facially neutral rules—a violation of employment equality legislation in most developed jurisdictions.
C3	Model-Based Reflex Agent	Observability: Partially Observable	Candidate information is frequently incomplete at application stage: employment gaps are unexplained, references are unavailable, and portfolio links may be broken. The MBRA fills these observational gaps using an internal model of the "ideal candidate" derived from historical hiring data—inferring missing information based on patterns in comparable past applications. Limitation: The ideal-candidate model is an artefact of historical hiring decisions. In most industries, those decisions reflected human biases, producing workforces disproportionately constituted by white, male graduates from elite institutions. The model's gap-filling assumptions systematically bias toward these demographic characteristics. Ethical Concern: When the model interprets an employment gap as low commitment rather than as potential evidence of caregiving responsibilities, disability, economic hardship, or immigration status challenges, it systematically disadvantages women, economically marginalised candidates, and candidates from immigrant backgrounds—reproducing historical discrimination through probabilistic inference.

#	Agent Type	Environment Characteristic	Analysis & Ethical Implications
C4	Model-Based Reflex Agent	Episodicity: Sequential	Recruitment proceeds through sequential stages: application screening, skills assessment, telephone interview, panel interview, and offer. The MBRA tracks each candidate's state across all stages, using prior performance assessments to inform evaluation at subsequent stages, creating a stateful model of each candidate's trajectory through the process. Limitation: Biases introduced at stage 1 are carried forward and amplified through all subsequent stages. A candidate who receives a marginal initial screening score is more likely to receive harsh evaluation at the interview stage, as interviewers are primed by the AI's prior assessment. The sequential structure transforms individual bias into systemic harm. Ethical Concern: Sequential AI-mediated recruitment is more dangerous than single-stage filtering because bias accumulates invisibly and becomes self-confirming. By the time human interviewers participate, they review a pre-filtered candidate pool that reflects algorithmic biases while believing they are selecting from the field's best—a particularly insidious form of discrimination in which human judgment is conscripted into algorithmic prejudice.
C5	Goal-Based Agent	Dynamism: Static	For a defined role with stable requirements and a bounded application pool, the GBA systematically plans toward a hiring goal, evaluating all candidates against all role requirements in a structured search for the optimal hire. In a static environment, this search can be thorough, consistent, and documented. Limitation: The hiring "goal" is defined by whoever designs the system, and this definition invariably encodes existing organisational values and assumptions. Goals defined in terms of "culture fit," "leadership potential," or "communications excellence" typically reflect the characteristics of the existing workforce, which in most organisations is not demographically representative.

#	Agent Type	Environment Characteristic	Analysis & Ethical Implications
			Ethical Concern: A goal-based agent faithfully optimises for whatever goal it is given, achieving discrimination with unprecedented efficiency and scale when that goal encodes historical biases. The attribution of discriminatory outcomes to the AI system rather than to human goal-setters also obscures accountability, making it structurally difficult for affected candidates to identify and legally challenge discriminatory practices.
C6	Goal-Based Agent	Determinism: Stochastic	Stochastic elements pervade hiring environments: vague job descriptions, inconsistent data quality across applications, unpredictable candidate presentation, and variable assessor interpretation. The GBA must pursue its hiring goal despite these uncertainties, developing contingent reasoning paths for different possible candidate scenarios. Limitation: When stochasticity prevents reliable merit-signal detection, the GBA falls back on statistical proxies—educational institution prestige, email domain provider, name etymology, residential postcode—all of which correlate strongly with race, gender, and socioeconomic status but not necessarily with job performance. Ethical Concern: Reliance on statistical proxies in stochastic environments is where algorithmic discrimination is most concentrated and most difficult to detect. The system produces merit-framed outputs while making demographic-based decisions—a form of disparate treatment that is both ethically indefensible and, under the Equality Act 2010 and analogous legislation, legally prohibited.
C7	Utility-Based Agent	Continuity: Continuous	Modern AI hiring platforms evaluate candidates on continuous scales: NLP-derived sentiment and confidence scores from cover letters, engagement metrics, eye-contact duration, and speech-rate analysis from video interviews, and composite predictive success indices from psychometric assessments. The UBA combines these into a

#	Agent Type	Environment Characteristic	Analysis & Ethical Implications
			weighted composite utility score and ranks candidates accordingly. Limitation: Continuous scoring creates false precision masking fundamental subjectivity. A difference of 0.4 utility score points between candidates is treated as meaningful by the system, yet this gap falls within measurement error and is entirely contingent on the arbitrary weighting of utility function components—weights chosen by system designers with no particular ethical accountability. Ethical Concern: Continuous utility evaluation in video interviews extends algorithmic surveillance into intimate personal domains—the candidate's home environment, vocal affect, facial microexpressions, and involuntary physiological responses. This constitutes an invasion of private life that pre-AI hiring processes could not achieve. It systematically disadvantages candidates with disabilities, neurodivergent individuals, and those whose cultural communication norms diverge from majority patterns encoded in the utility function.
C8	Utility-Based Agent	Determinism: Strategic	Contemporary hiring ecosystems are strategic: the AI hiring system, applicants using AI-assisted application tools, diversity compliance monitoring software, and human hiring managers all interact as utility-maximising agents. Each actor optimises its own objective, creating dynamic interactions that shape system-wide outcomes in ways that no single actor controls. Limitation: The strategic environment creates arms-race dynamics. Applicants learn to optimise applications for the AI's utility function through keyword stuffing and ATS-optimised formatting. This forces the AI to evolve its criteria, which generates an escalating competition that systematically advantages candidates with resources to access AI-coaching and professional optimisation services.

#	Agent Type	Environment Characteristic	Analysis & Ethical Implications
			Ethical Concern: Strategic hiring environments directly translate socioeconomic privilege into algorithmic advantage, independent of merit. The candidate from a privileged background with access to AI-powered application coaching is structurally advantaged over an equally qualified candidate who lacks those resources. This constitutes a form of algorithmic gatekeeping that reproduces and amplifies existing social inequalities under the guise of objective evaluation.
C9	Learning Agent	Dynamism: Dynamic	Labour markets evolve continuously: skills become obsolete, new roles emerge, and organisational requirements change with market conditions. A learning agent adapts its evaluation criteria in response, continuously updating its model of the "ideal candidate" based on new hiring outcomes and post-hire performance feedback. Limitation: The learning agent can only observe the performance of candidates it hired, never those it rejected. Its learning is constrained to a fundamentally non-representative sample. Without counterfactual data—the performance of rejected candidates if they had been hired—the system cannot determine whether its historical rejections were correct, and optimises only for replication of past decisions. Ethical Concern: A learning agent without explicit debiasing in a dynamic labour market converges toward precise optimisation of historical bias rather than toward fairness. The Amazon AI recruitment tool (Dastin, 2018) demonstrated this failure concretely: trained on male-dominated historical hiring outcomes, it learned to penalise resumes containing the word "women's," treating gender as a negative predictor of suitability.
C10	Learning Agent	Episodicity: Sequential	The learning agent improves its predictive accuracy across sequential hiring rounds, refining its model of candidate success based on cumulative feedback integrating data from application, assessment,

#	Agent Type	Environment Characteristic	Analysis & Ethical Implications
			interview, and post-hire performance phases. Over time, it develops increasingly sophisticated predictors of job success from this multi-stage evidence base. Limitation: Sequential learning requires performance feedback over extended time horizons—often one to three years—between the hiring decision and reliable performance assessment. In the absence of this delayed feedback, the system optimises for features that predicted past hiring decisions (observable) rather than actual job performance (unobservable at decision time). Ethical Concern: Learning from sequential outcomes without long-horizon causal feedback creates conditions for the algorithm to optimise spurious correlations that align with demographic characteristics. A candidate's university name predicts prior hires because prior hires graduated from those universities; the learning agent may incorporate this correlation without any mechanism to distinguish it from genuine merit signals—embedding discrimination in a "learning" wrapper that appears objective.

6.4 Societal Impact

AI-based recruitment systems have far-reaching societal implications that extend well beyond individual hiring decisions:

Reproduction of Structural Inequality: When AIHR systems are trained on historical hiring data that reflects decades of discriminatory practice, they learn to optimise for candidates who resemble past hires rather than for candidates who will perform best. This creates a closed loop in which past discrimination informs present algorithmic decision-making, which perpetuates future structural inequality. This dynamic has been documented empirically across multiple industries and jurisdictions.

Labour Market Access and Economic Opportunity: Access to employment is one of the most fundamental determinants of life outcomes—affecting income, housing security, healthcare access, educational opportunity for children, and social participation. When AIHR systems systematically disadvantage certain demographic groups, they are not merely making inefficient hiring decisions; they are shaping the distribution of life chances across society.

Opacity and the Impossibility of Challenge: Candidates who are rejected by AI hiring systems typically receive no explanation for the decision, have no access to the data or model that generated it, and have no meaningful mechanism for challenge. This opacity is not incidental—it is structural. Complex ML models cannot generate human-comprehensible explanations of individual decisions, and organisations are reluctant to expose their systems to scrutiny.

The Homogenisation of Talent: By optimising for candidates who resemble historical successful hires, AIHR systems actively discourage cognitive, experiential, and demographic diversity in organisations. Research consistently demonstrates that diverse teams outperform homogeneous ones on complex problem-solving tasks—meaning that algorithmic homogenisation of talent is not only unjust but economically counterproductive.

Psychological and Social Harm to Candidates: Being evaluated by an AI system on the basis of one's facial microexpressions, voice tone, and eye-contact patterns during a video interview constitutes a form of intrusive surveillance that many candidates experience as dehumanising. The psychological harm of being assessed by an opaque algorithmic system—and rejected without explanation—is a real and under-acknowledged social cost.

6.5 Ethical Concerns

- **Algorithmic Discrimination and Protected Characteristics:** Multiple jurisdictions' employment equality legislation prohibits discrimination on the basis of race, gender, age, disability, and other protected characteristics. AI systems that generate disparate impact against these groups—even without discriminatory intent—may violate these protections. However, enforcement of these protections against algorithmic systems remains legally and practically underdeveloped.
- **Training Data Quality and Historical Bias:** AI systems trained on historical hiring data learn the patterns of human decision-making embedded in that data—including

conscious and unconscious biases. Garbage in, garbage out is not merely a technical principle; it is an ethical one. The quality and representativeness of training data is a moral responsibility of the system's developers.

- **Explainability and the Right to Reasons:** Employment law in many jurisdictions provides candidates with a right to understand the reasons for adverse hiring decisions. Complex ML models cannot satisfy this right as currently deployed, creating a fundamental incompatibility between algorithmic decision-making and legal standards of procedural fairness.
- **Video Interview Analysis and Informed Consent:** Analysing candidates' facial expressions, voice tone, and physiological responses without their fully informed consent raises serious privacy concerns. Candidates who need employment are not in a position to freely refuse consent, making formal consent frameworks inadequate in practice.
- **The Accountability Gap:** When an AI system makes a discriminatory hiring decision, accountability is diffuse across developers, training data curators, system deployers, and human users of AI recommendations. This diffusion of responsibility creates a structural accountability gap in which no actor bears meaningful responsibility for discriminatory outcomes.

7. Legal and Ethical Issues Confronting Society Through AI

7.1 Privacy and Surveillance

The right to privacy—understood as the right to control information about oneself and to move through the world without being monitored—is a foundational element of liberal democratic societies, enshrined in Article 8 of the European Convention on Human Rights, the Fourth Amendment to the United States Constitution, and analogous provisions in constitutional systems worldwide. AI surveillance technologies challenge this right in ways that are qualitatively different from prior surveillance technologies.

Traditional surveillance was constrained by cost—maintaining human surveillance of an individual required significant resources, which meant that surveillance was practically limited to specific suspects under specific circumstances. AI surveillance removes this constraint. A single facial recognition network can monitor millions of individuals simultaneously at negligible marginal cost per person. This scalability transforms surveillance from an exceptional investigative tool into a permanent ambient condition of public life.

The legal frameworks designed to constrain surveillance were developed in a pre-AI context and are systematically inadequate for addressing AI-enabled surveillance at scale. Requirements for individualised suspicion, judicial authorisation, and proportionality—the traditional safeguards against surveillance overreach—are difficult to apply to systems that continuously scan entire populations. The European AI Act (2024) represents the most comprehensive attempt to regulate AI surveillance to date, classifying real-time biometric identification in public spaces as a "high-risk" AI application subject to strict requirements, and prohibiting certain uses categorically.

The concept of contextual integrity (Nissenbaum, 2004) provides a useful analytical lens: information flows appropriately when they match the norms of the context in which information was originally shared. When an individual appears on a public street, they accept that others present may observe them; they do not accept that their presence will be biometrically recorded, matched against a database, and retained indefinitely. AI surveillance violates contextual integrity not by observing what is visible, but by transforming that observation into a permanent, searchable, sharable record.

7.2 Bias and Discrimination

Algorithmic bias—the systematic, unjustified differential treatment of individuals based on demographic characteristics by AI systems—represents perhaps the most politically urgent ethical challenge in contemporary AI deployment. Its significance lies not merely in individual instances of unfair treatment, but in its capacity to entrench, amplify, and legitimise structural inequality at scale.

Bias in AI systems can arise from multiple sources across the system lifecycle:

- **Historical bias:** Training data reflects historical human decisions, which themselves embodied the biases of their era. An AI system trained to predict job success based on the characteristics of prior successful hires learns whatever demographic patterns are embedded in that history.
- **Representation bias:** Training datasets that under-represent certain demographic groups produce models that perform poorly on those groups—exactly the pattern observed in facial recognition accuracy disparities.
- **Measurement bias:** Proxy variables used as labels or features may correlate with protected characteristics. A resume's first name, university name, or email domain can serve as a proxy for race or socioeconomic status.
- **Aggregation bias:** Models trained on aggregate population data may not appropriately represent the heterogeneity of subgroups, producing accurate average performance and systematically poor performance for minorities.
- **Deployment bias:** Systems deployed in contexts different from those for which they were designed may perform in unexpected and harmful ways.

The consequences of algorithmic bias are not distributed neutrally—they disproportionately burden already-marginalised communities. This is not coincidental; it reflects the fact that AI systems are typically designed by and for dominant demographic groups, trained on data generated by historical practices that disadvantaged minorities, and deployed in contexts where those minorities have the least power to challenge adverse decisions.

Legal frameworks for addressing algorithmic discrimination are developing but remain inadequate. In the United States, Title VII of the Civil Rights Act prohibits employment

discrimination based on race, sex, and other protected characteristics, and has been interpreted to cover disparate impact—disproportionate adverse effects on protected groups—as well as disparate treatment. The EU Artificial Intelligence Act designates AI hiring systems as high-risk and imposes requirements for transparency, human oversight, and fundamental rights impact assessment. The UK Equality Act 2010 similarly prohibits indirect discrimination in employment.

However, enforcement of these frameworks against algorithmic systems faces significant practical barriers: the opacity of ML models makes disparate impact difficult to detect and prove; the diffusion of responsibility across developers, deployers, and users makes accountability assignment legally complex; and the pace of AI deployment consistently outstrips the capacity of regulatory institutions to respond.

7.3 The Role of Human Judgment

The question of human judgment in AI-mediated decision-making is philosophically complex and practically urgent. The use of AI in high-stakes decisions—criminal justice, employment, healthcare, immigration—raises fundamental questions about the appropriate division of cognitive labour between humans and machines, the nature of accountability, and the irreducible value of human judgment in decisions that affect human dignity.

Proponents of AI decision support argue that human judgment is systematically unreliable: it is subject to cognitive biases, fatigue, emotional states, and unconscious prejudices that produce arbitrary and inconsistent decisions. On this view, algorithmic decision-making is not a threat to justice but a potential remedy for the inconsistencies of human judgment.

This argument is important but insufficient for three reasons:

First, algorithmic and human biases differ in kind, not merely in degree. Human biases are inconsistent, vary across individuals, and are partially self-correcting through social feedback and deliberation. Algorithmic biases are consistent, scale across millions of decisions simultaneously, and are self-reinforcing when baked into systems that generate their own training data. A biased human recruiter affects the candidates they personally evaluate; a biased algorithmic system affects every candidate processed by every organisation using that system.

Second, the claim to algorithmic objectivity is itself ideologically loaded. The choice of training data, features, utility functions, performance metrics, and optimization objectives are all human decisions that encode human values. Presenting these value-laden choices in the language of mathematics creates a false appearance of value-neutrality—what Mittelstadt et al. (2016) call the "mathematical veneer" of algorithmic decision-making. This veneer is dangerous because it insulates power-laden choices from democratic scrutiny.

Third, certain decisions are irreducibly human in character. Decisions that fundamentally affect human dignity, autonomy, and opportunity—who gets a job, who is stopped by the police, who receives bail—carry moral weight that cannot be delegated to algorithms without loss. These decisions involve not merely the assessment of facts but the exercise of moral judgment: balancing competing values, recognising the unique circumstances of individuals, and accepting responsibility for the consequences of choices. Algorithmic systems cannot accept moral responsibility; they can only produce outputs.

The appropriate response to the tension between human and algorithmic judgment is not a binary choice between full automation and full human discretion, but a principled framework for human-AI collaboration that preserves human accountability while leveraging algorithmic efficiency. Key principles include: meaningful human review of all high-stakes AI decisions; explainability requirements that enable humans to understand and challenge algorithmic recommendations; regular bias audits with mandatory remediation; and genuine democratic accountability for the design and governance of AI systems affecting the public interest.

The emerging concept of "meaningful human control" (Roff & Danks, 2018) is particularly relevant: human oversight of AI systems is meaningful only when the reviewing human has sufficient time, information, competence, and authority to override the system's recommendations. Rubber-stamp review—in which humans formally approve algorithmic decisions without the capacity or incentive to challenge them—provides the appearance of accountability without its substance.

8. Conclusion

This report has examined the ethical and societal implications of AI engineering solutions through a structured analysis of two high-stakes applications: Facial Recognition and AI Surveillance, and AI-Based Recruitment and Hiring. The analysis across five agent types, five environment categories, and twenty-five possible combinations has demonstrated that the ethical character of an AI system is not separable from its technical architecture—the choice of agent type, the nature of the deployment environment, the design of the utility function, and the sources of training data are all moral choices with consequential implications for individual rights and social justice.

Several overarching conclusions emerge from this analysis:

- AI systems do not merely reflect the world; they reshape it. Systems trained on historical data reproduce historical inequalities with unprecedented efficiency and scale. This is not a technical bug—it is a structural feature of learning from the past in unjust societies.
- The promise of algorithmic objectivity is largely illusory. Every AI system embeds human value choices at multiple points in its design and deployment. The appropriate response is not to abandon these choices to mathematics but to make them explicit, accountable, and democratically contestable.
- The harms of AI are not evenly distributed. They concentrate on communities that already face structural disadvantage—racial minorities, women, economically marginalised groups—amplifying existing inequalities rather than transcending them.
- Technical and ethical sophistication must advance together. More capable AI agents—utility-based systems and learning agents—have greater potential for both benefit and harm. The complexity of these systems demands correspondingly sophisticated ethical governance, not merely more capable engineering.
- Regulatory frameworks are necessary but not sufficient. The EU AI Act, proposed biometric surveillance legislation, and equal employment opportunity law create important constraints, but enforcement challenges, the pace of technological development, and the global nature of AI deployment limit the effectiveness of any single regulatory jurisdiction.

The engineering community bears a special responsibility in this context. Engineers are not merely technical executors of others' decisions—they are co-designers of the systems that shape social outcomes. Fulfilling this responsibility requires cultivating the capacity for ethical reasoning, engaging with affected communities, designing for transparency and accountability, and advocating for governance frameworks that protect human dignity in the age of intelligent machines.

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Appendix A – Agent–Environment Combination Matrix

The following matrices show the assignment's 25-cell pedagogical mapping. Rows are the five agent architectures. Columns are the five environment dimensions. Cells marked ★ are the ten combinations discussed in the detailed analysis for each application. The star indicates a selected scenario for analysis, not a claim that the combination is the only possible design.

A.1 Facial Recognition and AI Surveillance :

Agent Type	Observability	Determinism	Episodicity	Dynamism	Continuity
Simple Reflex Agent (SRA)	★ C1	★ C2	—	—	—
Model-Based Reflex Agent (MBRA)	★ C3	—	—	★ C4	—
Goal-Based Agent (GBA)	—	★ C6	★ C5	—	—
Utility-Based Agent (UBA)	★ C8	—	—	—	★ C7
Learning Agent (LA)	—	—	—	★ C9	★ C10

A.2 AI-Based Recruitment and Hiring :

Agent Type	Observability	Determinism	Episodicity	Dynamism	Continuity
Simple Reflex Agent (SRA)	★ C2	—	★ C1	—	—
Model-Based Reflex Agent (MBRA)	★ C3	—	★ C4	—	—
Goal-Based Agent (GBA)	—	★ C6	—	★ C5	—
Utility-Based Agent (UBA)	—	★ C8	—	—	★ C7
Learning Agent (LA)	—	—	★ C10	★ C9	—